

André Freitas

PhD student in Physics & Computer Science

andre.marreiros.freitas@gmail.com · andremfreitas.github.io · github.com/andremfreitas

EDUCATION

University of Rome “Tor Vergata” & Institut Polytechnique de Paris 2023–present
PhD Physics & CS Italy / France

- Marie Skłodowska-Curie doctoral network fellow.
- Thesis: *Large eddy simulation models in a deep machine learning loop.*

Delft University of Technology 2021–2023
MSc Aerospace Engineering Netherlands

- Thesis: *Numerical study on turbulence induced vibrations of fuel rods using an Anisotropic Pressure Fluctuations Model.*
- Electives focused on numerical analysis, high performance computing and machine learning.

Instituto Superior Técnico 2018–2021
BSc Aerospace Engineering Portugal
Electives in computational mathematics and control theory. No final year project.

EXPERIENCE

Summer 2026	Visiting Grad Student	Stanford University	USA
Summer 2026	Visiting Grad Student	Los Alamos National Laboratory	USA
Fall 2025	PhD Intern	NVIDIA & Forschungszentrum Jülich	Germany
Summer 2025	Visiting Grad Student	Los Alamos National Laboratory	USA
2022–2023	Research Intern	Nuclear Research Group (NRG)	Netherlands
Summer 2022	Aerodynamics Engineer Intern	PLANA	South Korea
2022	Teaching Assistant	TU Delft, Aerospace Engineering	Netherlands
2021	Undergrad Research Assistant	Instituto Superior Técnico	Portugal

RESEARCH INTERESTS

Statistical physics, multiscale systems modelling and control, differentiable programming for physics-informed machine learning, and high-performance solvers for large-scale GPU clusters. More recently, graph neural networks and stochastic interpolants.

PUBLICATIONS

Published

- A. Freitas**, K. Um, M. Desbrun, G. Eyink, A.A. Mailybaev, L. Biferale. On the importance of stochasticity in closures of turbulence. *Europhysics Letters* (2026).
- A. Freitas**, K. Um, M. Desbrun, M. Buzzicotti, L. Biferale. A posteriori closure of turbulence models: are symmetries preserved? *European Journal of Mechanics - B/Fluids* (2026).
- X. de Wit, **A. Freitas**, F. Toschi. Dynamics of small bubbles in non-dilute conditions. *Physical Review Fluids* (2026).
- A. Freitas**, K. Um, M. Desbrun, M. Buzzicotti, L. Biferale. Solver-in-the-loop approach to closure of shell models of turbulence. *Physical Review Fluids* (2025).

K. Zwijsen, **A. Freitas**, S. Tajfirooz, E.M.A. Frederix, A.H. van Zuijlen. An optimized anisotropic pressure fluctuation model for turbulence-induced vibrations. *Physics of Fluids* (2024).

In Review

A. Freitas, X. de Wit, Z. Wang, L. Biferale, F. Toschi. Statistical properties of turbulence under a smart Lagrangian forcing. (2025).

A. Freitas, X.M. de Wit, A. Gabbana, M. Woodward, F. Toschi, Y.T. Lin, D. Livescu. Learning turbulent transport via Mori–Zwanzig graph neural networks. (2026).

PRESENTATIONS

Learning turbulent transport via Mori–Zwanzig graph neural networks. 4th Bilateral Smart-Turb Smart-Heart Workshop, Sperlonga (2026).

Intermittency suppression via forced light particles. Chiral Particles and Odd Fluids, Eindhoven (2026).

Eulerian and Lagrangian turbulence: closure and control problems. 3rd Bilateral Smart-Turb Smart-Heart Workshop, Lecce (2025).

Statistical properties of turbulence under a smart Lagrangian forcing. APS Division of Fluid Dynamics Meeting, Houston (2025).

Statistical properties of turbulence under a smart Lagrangian forcing. 2nd European Fluid Dynamics Conference, Dublin (2025).

Differentiable LES of Rayleigh–Bénard convection. Euromech Colloquium “Data-driven fluid mechanics”, London (2025).

Solver-in-the-loop approach to turbulence closure. 1st European Fluid Dynamics Conference, Aachen (2024).

The Adjoint Method in Scientific Machine Learning. Complex Flows and Complex Fluids Seminar, Rome (2024).

GRANTS & FELLOWSHIPS

2026	Center of Turbulence Research (CTR) summer fellow	Stanford University
2023–2026	Marie Skłodowska-Curie Actions PhD Fellowship	European Union
2021	Research Initiation Grant	Instituto Superior Técnico

REFERENCES

Luca Biferale	biferale@gmail.com
Mathieu Desbrun	mathieu@cms.caltech.edu
Daniel Livescu	livescu@lanl.gov